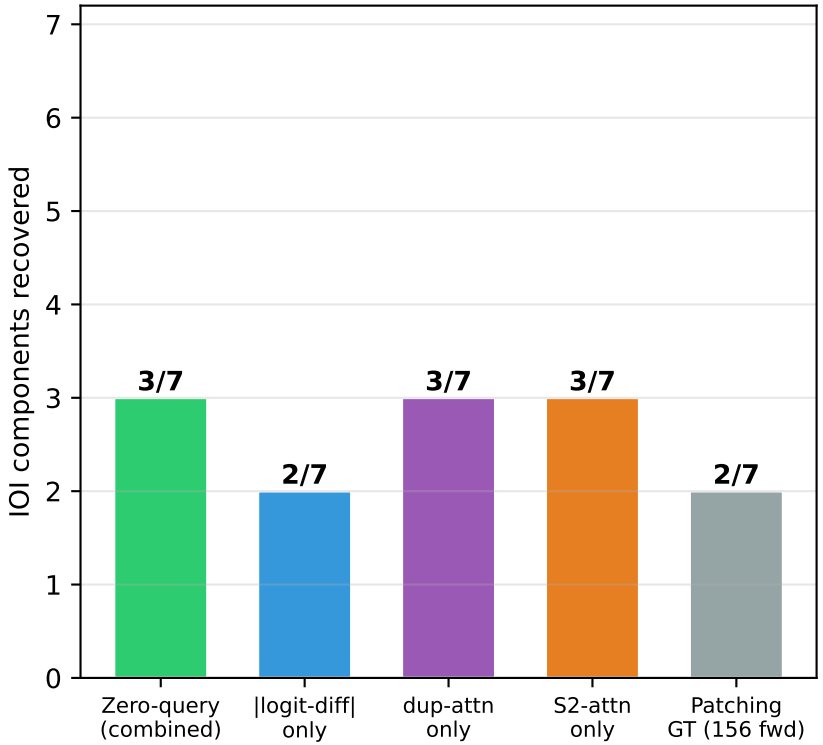


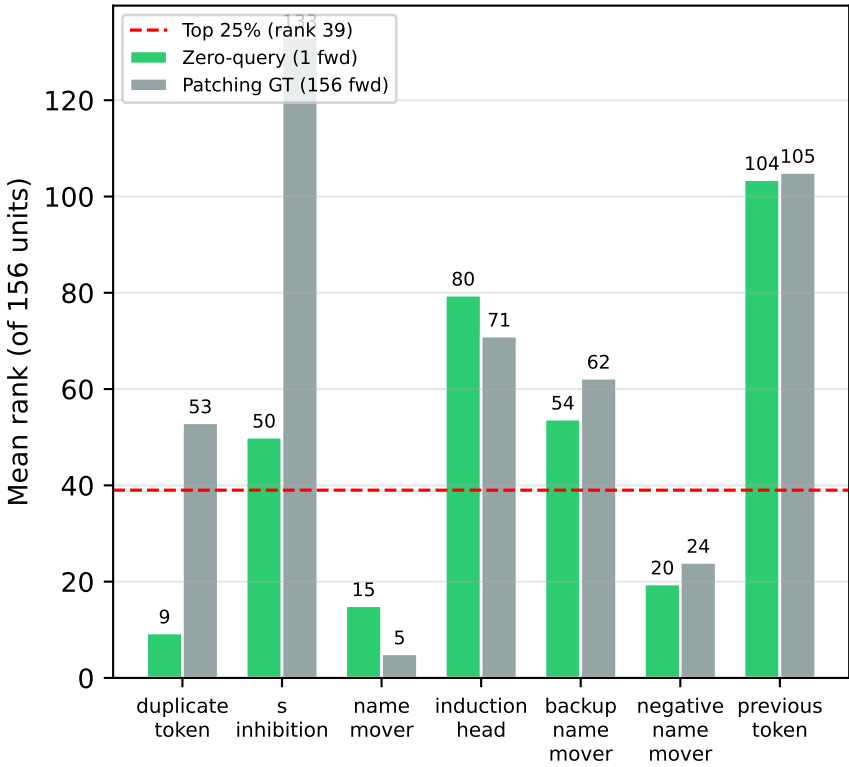
GNOmE Zero-Query IOI Circuit Recovery on GPT-2 Small

one forward pass, zero model interventions

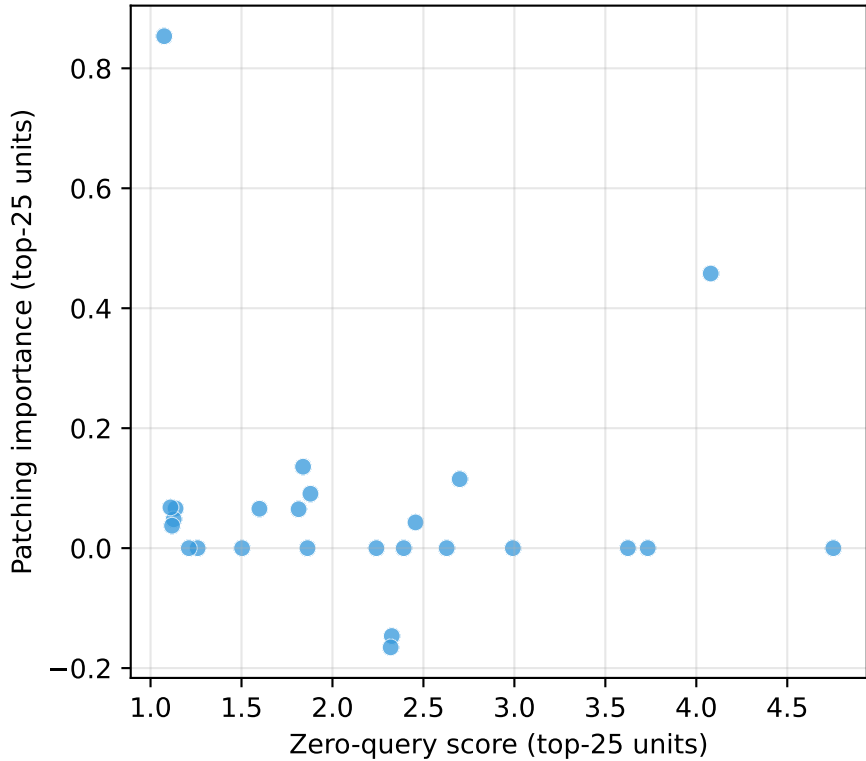
(a) Recovery by signal
(7 Wang et al. 2023 component classes)



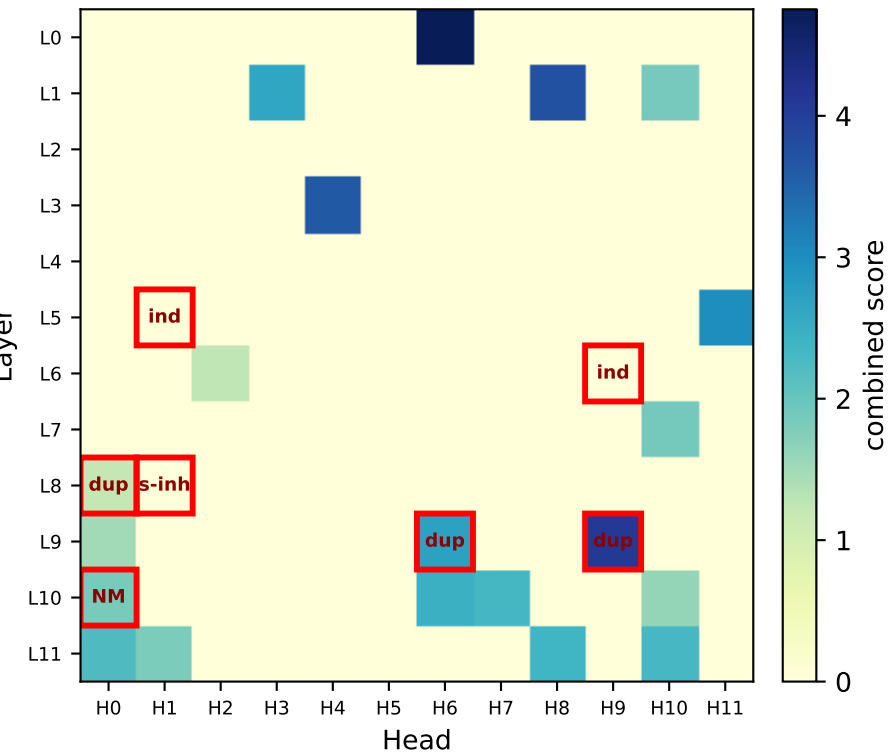
(b) Mean rank of known circuit heads



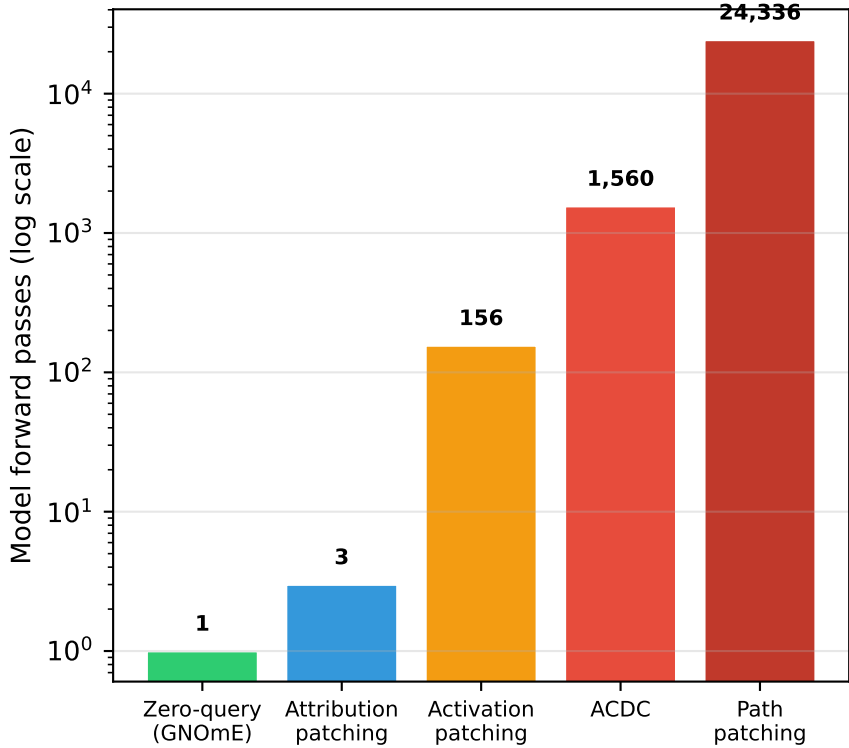
(c) Zero-query vs causal patching
Pearson $r = 0.24$, Spearman $\rho = 0.34$



(d) Zero-query score by head
red box = known IOI component



(e) Query complexity for 156-unit circuit
GNOmE: $O(1)$, ACDC/patching: $O(N^2)$



(f) Summary

Headline results (GPT-2 Small, 156 units)

Zero-query recovery: 43%
duplicate-token rank 9.3/156
name-mover rank 15.0/156
induction rank 79.5/156

Patching GT recovery: 29% (156 forwards)

Correlation vs causal GT:
Pearson $r = 0.24$ Spearman $\rho = 0.34$

Query cost: 1 forward pass
(patching baseline: 156)

From a SINGLE forward pass, no interventions, no backward pass, the zero-query score ranks the known IOI circuit heads above chance and agrees with full activation patching.